THE ERASURE ENTROPY OF A FORCED COORDINATE UNDER EVERY BIAS

YITZCHAK SHMALO

Abstract. A monotone Boolean function $F_{k,t}$, built from a majority function on k inputs with one hidden coordinate x by substituting threshold functions of $2t$ fresh variables, has x as the only variable lying both in a shortest minterm and in a shortest maxterm. The ratio E_x/Inf_x of the erasure entropy to the influence at x was known to be unbounded when k and t grow together, and to grow linearly in k for $t = 1$. We show that for every fixed $t \geq 1$ it equals $(1 + o(1)) \log_2(1/\mathrm{Inf}_x) = (1 + o(1))(k - 1) \log_2(1/P_t)$, where P_t is the probability that a binomial variable with parameters $2t$ and $\frac{1}{2}$ is at least t. More generally the same holds for the underlying majority function under every rational product bias $P \in (\frac{1}{2}, 1)$, the phase of the relevant coefficients advancing by $\arccos \frac{2P-1}{2P}$, an irrational multiple of π. At $P = \frac{1}{2}$ that angle is $\frac{\pi}{2}$, and the behaviour changes: there the coefficients are values of Krawtchouk polynomials at the middle of their range, half of them vanish, and the ratio is $\frac{1}{2} \log_2(1/\mathrm{Inf}_x) - \frac{1}{2} \log_2 \log_2(1/\mathrm{Inf}_x) + c + O(1/\log_2(1/\mathrm{Inf}_x))$ with $c = \frac{1}{2} + \frac{3}{4} \log_2 e - \frac{1}{2} \log_2 \frac{\pi}{2}$, which we derive from an exact single-sum formula.

1. Introduction

For $f\colon \{0,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$ and let U be a random subset with $\Pr[U = S] = \hat{f}(S)^2$. The Fourier entropy-influence conjecture of Friedgut and Kalai [7] asks for an absolute C with $H(U) \leq C\,\mathbb{E}|U|$. It is known for symmetric functions and read-once decision trees [16], it is preserved under suitable compositions [15], and weaker forms and bounds with extra factors are in [4, 19, 11, 8]; the min-entropy version is known for read-k DNFs with k fixed [2, 1]; lower bounds on the best constant are in [10]. The erasure entropy of f at a coordinate x is $E_x(f) = H(\mathbf{1}[x \in U] \mid U \setminus \{x\})$, the conditional entropy of one coordinate given the others in the sense of Verdú and Weissman [18], and with $\mathrm{Inf}_x(f) = \Pr[x \in U]$ the conjecture holds if and only if $\sup_f \min_{x:\mathrm{Inf}_x(f)>0} E_x(f)/\mathrm{Inf}_x(f)$ is finite [22], which builds on the reduction of González, MacManus and Pereyra [6] to a statement about one coordinate. Since conditioning does not increase entropy,

$$\frac{E_x(f)}{\mathrm{Inf}_x(f)} \leq \frac{h_2(\mathrm{Inf}_x(f))}{\mathrm{Inf}_x(f)} \leq \log_2 \frac{e}{\mathrm{Inf}_x(f)} \tag{1}$$

for every f and x with $\mathrm{Inf}_x(f) > 0$.

The question is which coordinate to take. In [23] several selection rules are shown to fail on monotone functions, among them the rule that takes, among the variables lying both in a shortest minterm and in a shortest maxterm, one of smallest ratio. Fix an odd $k = 2m + 1$, let $y^0 \in \{0,1\}^k$ have $y^0_j = 1$ for $j \leq m$ and $y^0_j = 0$ for $j > m$, let $y^1 = y^0 + e_{m+1}$, let M be MAJ_k with the point y^1 removed from its true set, and

$$\tilde{f}_k(y, x) = M(y) \vee \big(x \wedge [y \geq y^0]\big).$$

For $t \geq 1$ the function $F_{k,t}$ of [23, Section 5] replaces y_j by the threshold function "at least t of $2t$" of a block of fresh variables for $j \leq m$, and by "at least $t + 1$ of $2t$" for $j > m$. It is monotone, and x is the only variable lying both in a shortest minterm and in a shortest maxterm. Moreover [23, proof of Proposition 12] $E_x(F_{k,t})$ and $\mathrm{Inf}_x(F_{k,t})$ equal the erasure entropy and the

influence of x in the expansion of $\tilde{f}_k$ in the μ-biased Fourier basis [14, Chapter 8] of the product measure with $\Pr[y_j = 1] = P_t$ for $j \le m$ and $1 - P_t$ for $j > m$, where

$$P_t = \Pr\big[\mathrm{Bin}(2t, \tfrac{1}{2}) \ge t\big] = \frac{1}{2} + \binom{2t}{t} 2^{-2t-1};$$

the biased form of the conjecture is in [12], and a sharp lower bound on the biased Fourier entropy by a sum of coordinate-wise terms is in [3]. For $t = 1$, $P_1 = \frac{3}{4}$ and the ratio at x is $(1 + o(1))(k-1)\log_2 \frac{4}{3}$ [24]. Here we treat every fixed $t \ge 1$, and more generally every rational bias.

For $P \in (0,1)$ write $E_x^P(\tilde{f}_k)$ and $\mathrm{Inf}_x^P(\tilde{f}_k)$ for the erasure entropy and the influence of x for the law $c(S)^2$, where $c(S)$ are the coefficients of $\tilde{f}_k$, in ± 1 values, in the orthonormal basis of the product measure with $\Pr[y_j = 1] = P$ for $j \le m$, $\Pr[y_j = 1] = 1 - P$ for $j > m$, and x uniform. At $P = \frac{1}{2}$ these are $E_x(\tilde{f}_k)$ and $\mathrm{Inf}_x(\tilde{f}_k)$. Let $\alpha_P = \arccos \frac{2P-1}{2P}$.

Theorem 1. *Let $P \in (\frac{1}{2}, 1)$ with α_P/π irrational. Then $\mathrm{Inf}_x^P(\tilde{f}_k) = P^{k-1}$ and, as $k \to \infty$ through odd values,*

$$\frac{E_x^P(\tilde{f}_k)}{\mathrm{Inf}_x^P(\tilde{f}_k)} = (1 + o(1)) \log_2 \frac{1}{\mathrm{Inf}_x^P(\tilde{f}_k)} = (1 + o(1))\,(k-1) \log_2 \frac{1}{P}.$$

The condition holds for every rational $P \in (\frac{1}{2}, 1)$.

Corollary 2. *For every fixed $t \ge 1$, $\mathrm{Inf}_x(F_{k,t}) = P_t^{k-1}$ and $E_x(F_{k,t})/\mathrm{Inf}_x(F_{k,t}) = (1+o(1))(k-1)\log_2(1/P_t)$ as $k \to \infty$. In particular the minterm-maxterm rule of* [23] *has unbounded ratio on the functions $F_{k,t}$ with t fixed.*

The corollary is Theorem 1 with $P = P_t$, which is rational, together with the reduction quoted above. The case $P = \frac{3}{4}$ is the theorem of [24], and the proof below follows that paper; what makes it work for every P are three identities on the unit circle, (5) below.

At $P = \frac{1}{2}$ the rotation angle $\alpha_{1/2}$ is $\frac{\pi}{2}$ and the argument breaks down for a reason that can be made exact. Write $\Psi(r) = r^2 \log_2(1 + r^{-2}) + \log_2(1 + r^2)$ for $r > 0$ and $\Psi(0) = 0$.

Theorem 3. *Let $k = 2m + 1$. Then $\mathrm{Inf}_x(\tilde{f}_k) = 2^{-2m}$ and*

$$\frac{E_x(\tilde{f}_k)}{\mathrm{Inf}_x(\tilde{f}_k)} = 4^{-m} \sum_{b=0}^{m-1} \binom{2m}{2b+1} \Psi\Big(\binom{m}{b}\binom{2m}{m} \Big/ \binom{2m}{2b}\Big).$$

Moreover $\big|E_x(\tilde{f}_k)/\mathrm{Inf}_x(\tilde{f}_k) - m + \frac{1}{2}\log_2 m\big| \le C$ for an absolute constant C, and

$$\frac{E_x(\tilde{f}_k)}{\mathrm{Inf}_x(\tilde{f}_k)} = m - \frac{1}{2}\log_2 m + c_0 + O\Big(\frac{1}{m}\Big), \qquad c_0 = \frac{3}{4}\log_2 e - \frac{1}{2}\log_2 \frac{\pi}{2} = 0.75627\ldots,$$

so that, with $L = \log_2(1/\mathrm{Inf}_x(\tilde{f}_k)) = 2m$, the ratio is $\frac{1}{2}L - \frac{1}{2}\log_2 L + c_0 + \frac{1}{2} + O(1/L)$.

In [23, Proposition 11] the ratio of $\tilde{f}_k$ was shown to be at least $(\frac{1}{8} - o(1))(k-1)$; the theorem gives the constant $\frac{1}{2}$. The factor $\frac{1}{2}$ has a direct meaning: on exactly half of the spectral weight of the derivative in x the average over x has a vanishing coefficient, so there $x \in U$ is determined by $U \setminus \{x\}$ and contributes no entropy. As $t \to \infty$ with k fixed the ratio of $F_{k,t}$ tends to that of $\tilde{f}_k$ [23, Proposition 12], while for each fixed t the ratio is asymptotic to the bound (1); the two limits do not commute.

2. The ratio as a double sum

All logarithms are to base 2 unless written ln. For $p, q \geq 0$ let $\Phi(p,q) = p \log \frac{p+q}{p} + q \log \frac{p+q}{q}$, with $\Phi(p,0) = \Phi(0,q) = 0$, so that $\Psi(r) = \Phi(r^2, 1)$. Then $\Psi \geq 0$, $\Psi(r) \geq \log r^2$ for $r \geq 1$, and $\Psi(r) \leq \log r^2 + 1 + \log e$ for $r \geq 1$.

Fix $P \in [\frac{1}{2}, 1)$ and put $q = 1 - P$ and $\sigma = \sqrt{Pq}$. In the basis $\prod_{j \in S} \phi_j(y_j)$, with $\phi_j(y) = (y - \mu_j)/\sigma$, $\mu_j = P$ for $j \leq m$ and $\mu_j = q$ for $j > m$, and with $+1$ for true, $\tilde{f}_k = \bar{A} + (2x - 1)D$, where

$$D = \mathbf{1}[y = y^0] + \mathbf{1}[y = y^1], \qquad \bar{A} = 2N - 1 + \mathbf{1}[y = y^0] - \mathbf{1}[y = y^1], \qquad N = \mathbf{1}[|y| \geq m + 1].$$

For $S \subseteq [k]$ let $u = |S \cap \{1, \ldots, m\}|$ and $v = |S \cap \{m+2, \ldots, k\}|$, and put

$$X(z) = q + Pz, \qquad Y(z) = P + qz,$$

$$H_{uv}(z) = (z-1)^{u+v} X(z)^{m-u} Y(z)^{m+1-v}, \qquad t_{uv} = \sum_{c \geq m+1} [z^c] H_{uv}(z).$$

Lemma 4. *If $m + 1 \in S$ then $\hat{D}(S) = 0$. If $m + 1 \notin S$ then $\hat{D}(S) = (-1)^v \sigma^{u+v} P^{2m-u-v}$ and $\hat{\bar{A}}(S) = 2\sigma^{u+v} t_{uv} - \mathbf{1}[S = \emptyset] + (P - q)\hat{D}(S)$.*

Proof. This is [24, Lemma 1] with P in place of $\frac{3}{4}$. For a point w, $\widehat{\mathbf{1}[y = w]}(S) = \Pr[y = w] \prod_{j \in S} \phi_j(w_j)$, and $\Pr[y = y^0] = P^{2m+1}$, $\Pr[y = y^1] = P^{2m} q$. If $m + 1 \in S$ the two terms of $\hat{D}(S)$ are $P^{2m+1}\phi_{m+1}(0) = -P^{2m}\sigma$ and $P^{2m} q\, \phi_{m+1}(1) = P^{2m}\sigma$ times the same product, and cancel. Otherwise both points give the product $(-1)^v (q/P)^{(u+v)/2}$, since $\phi_j(1) = \sqrt{q/P}$ for $j \leq m$ and $\phi_j(0) = -\sqrt{q/P}$ for $j \geq m + 2$; this gives $\hat{D}(S)$, and the difference of the two indicators has coefficient $(P - q)\hat{D}(S)$. Finally $\mathbb{E}[z^{y_j} \phi_j(y_j)] = \sigma(z - 1)$ and $\mathbb{E}[z^{y_j}] = 1 - \mu_j + \mu_j z$, so $\mathbb{E}[z^{|y|} \prod_{j \in S} \phi_j(y_j)] = \sigma^{u+v} H_{uv}(z)$ when $m + 1 \notin S$, and $\hat{N}(S)$ is the sum of its coefficients at z^c, $c \geq m + 1$. □

Let $\beta_u = \binom{m}{u} q^u P^{m-u}$ and, for $0 \leq u, v \leq m$,

$$r_{uv} = \frac{\left|2t_{uv} - \mathbf{1}[u = v = 0] + (P - q)(-1)^v P^{2m-u-v}\right|}{P^{2m-u-v}}.$$

Lemma 5. $\mathrm{Inf}_x^P(\tilde{f}_k) = P^{2m}$ *and* $E_x^P(\tilde{f}_k)/\mathrm{Inf}_x^P(\tilde{f}_k) = \sum_{u,v=0}^m \beta_u \beta_v \Psi(r_{uv})$.

Proof. As in [24, Lemma 2]: $E_x = \sum_S \Phi(\hat{\bar{A}}(S)^2, \hat{D}(S)^2)$ and $\mathrm{Inf}_x = \sum_S \hat{D}(S)^2$, the terms with $m + 1 \in S$ vanish, there are $\binom{m}{u}\binom{m}{v}$ sets with given (u, v) and $m + 1 \notin S$, $\sum_{u,v} \binom{m}{u}\binom{m}{v}(Pq)^{u+v} P^{2(2m-u-v)} = (Pq + P^2)^{2m} = P^{2m}$, and $\Phi(\hat{\bar{A}}^2, \hat{D}^2) = \hat{D}^2 \Psi(|\hat{\bar{A}}|/|\hat{D}|)$ with $|\hat{\bar{A}}(S)|/|\hat{D}(S)| = r_{uv}$. □

If $u + v \geq 1$ then $Q_{uv} = H_{uv}/(z - 1)$ is a polynomial and $t_{uv} = [z^m] Q_{uv}$: writing $Q_{uv} = \sum_c g_c z^c$, the sum $\sum_{c \geq m+1} (g_{c-1} - g_c)$ telescopes to g_m [24, Lemma 3].

3. The unbiased case

Here $P = q = \frac{1}{2}$, so $X = Y = (1 + z)/2$, the term $(P - q)\hat{D}$ is absent, and $\beta_u = \binom{m}{u} 2^{-m}$. For $0 \leq a \leq 2m$ let

$$\kappa_a = [z^m](1 - z)^a (1 + z)^{2m-a},$$

the value at a of the Krawtchouk polynomial of degree m for length $2m$ [13, Chapter 5, §7].

Lemma 6. $\kappa_a = 0$ *for odd a, and* $\kappa_{2b} = (-1)^b \binom{m}{b}\binom{2m}{m} / \binom{2m}{2b}$.

Proof. For a variable s,

$$\sum_{a=0}^{2m}\binom{2m}{a}\kappa_a s^a=[z^m]\big((1+z)+s(1-z)\big)^{2m}=[z^m]\big((1+s)+(1-s)z\big)^{2m}=\binom{2m}{m}(1-s^2)^m.$$

Comparing coefficients of s^a gives the claim. □

Proof of Theorem 3. By Lemma 5, $\mathrm{Inf}_x=2^{-2m}$. For $u+v\ge 1$ put $a=u+v-1$; then

$$\frac{2t_{uv}}{P^{2m-u-v}}=2^{2m-u-v+1}\,[z^m](z-1)^a\Big(\frac{1+z}{2}\Big)^{2m-a}=(-1)^a\kappa_a,$$

so $r_{uv}=|\kappa_{u+v-1}|$. For $u=v=0$, $t_{00}=\Pr[|y|\ge m+1]=\frac12$ and $r_{00}=0$. The weight of the pairs with $u+v=s$ is $\sum_{u+v=s}\beta_u\beta_v=\binom{2m}{s}4^{-m}$, and by Lemma 6 only odd $s=2b+1$ give $r\neq 0$, with $r=\binom{m}{b}\binom{2m}{m}/\binom{2m}{2b}$. This is the formula. Since $\binom{2m}{2b}\le\binom{2m}{m}$, these values satisfy $r\ge\binom{m}{b}\ge 1$.

For the asymptotics write $w_b=\binom{2m}{2b+1}4^{-m}$ and $\delta_b=(2b-m)^2/m$. The odd binomial coefficients of order $2m$ sum to 2^{2m-1}, so $\sum_b w_b=\frac12$, and if s is binomial with parameters $2m$ and $\frac12$ then $\sum_b w_b\delta_b\le\mathbb{E}[(s-1-m)^2]/m=\frac12+\frac1m$. We use $h(p)\ge 4p(1-p)=1-(2p-1)^2$ for the binary entropy h and, for $1\le j\le n-1$,

$$\frac{2^{nh(j/n)}}{\sqrt{8j(n-j)/n}}\le\binom{n}{j}\le\frac{2^{nh(j/n)}}{\sqrt{2\pi j(n-j)/n}}\tag{2}$$

[13, Chapter 10, Lemma 7], together with $\binom{2m}{m}\le 4^m/\sqrt{\pi m}$.

Lower bound. By (2) with $n=m$ and $8b(m-b)/m\le 2m$, $\log\binom{m}{b}\ge m\,h(b/m)-\frac12\log(2m)\ge m-\delta_b-\frac12\log m-\frac12$ for $1\le b\le m-1$, and this also holds for $b=0$. As $\Psi(r)\ge\log r^2$,

$$\frac{E_x}{\mathrm{Inf}_x}\ge 2\sum_b w_b\log\binom{m}{b}\ge m-\tfrac12\log m-\tfrac32-\tfrac2m.$$

Upper bound. For $1\le b\le m-1$, (2) applied to $\binom{m}{b}$ from above and to $\binom{2m}{2b}$ from below gives

$$\begin{aligned}\log r=\log\binom{m}{b}+\log\binom{2m}{m}-\log\binom{2m}{2b}&\le 2m-m\,h(b/m)-\tfrac12\log m+\tfrac32-\log\pi\\&\le m+\delta_b-\tfrac12\log m+1,\end{aligned}$$

the factors $b(m-b)/m$ cancelling. For $b=0$, $r=\binom{2m}{m}\le 4^m$ and $w_0=2m4^{-m}$. With $\Psi(r)\le\log r^2+1+\log e$,

$$\frac{E_x}{\mathrm{Inf}_x}\le\sum_{b\ge1}w_b\big(2m+2\delta_b-\log m+3+\log e\big)+w_0(4m+3)\le m-\tfrac12\log m+C.$$

Finally $\log_2(1/\mathrm{Inf}_x)=2m$, and $\frac12\log m=\frac12\log(2m)-\frac12$.

The constant. Let $m\ge 2$, write $z_b=2b-m$, so that $\delta_b=z_b^2/m$, and let C denote absolute constants. First, r is large for every b: since $i\mapsto\binom{m}{i}$ is log-concave, each term of $\binom{2m}{2b}=\sum_i\binom{m}{i}\binom{m}{2b-i}$ is at most $\binom{m}{b}^2$, so $\binom{2m}{2b}\le(m+1)\binom{m}{b}^2$ and, with $\binom{2m}{m}\ge 4^m/(2\sqrt{m})$ and $\binom{m}{b}\le 2^m$,

$$r=\binom{m}{b}\binom{2m}{m}\Big/\binom{2m}{2b}\ge\frac{\binom{2m}{m}}{(m+1)\binom{m}{b}}\ge\frac{2^m}{2(m+1)\sqrt{m}}.$$

For $r \geq 1$ and $u = r^{-2}$, $\Psi(r) - \log r^2 - \log e = \log e \left(\frac{\ln(1+u)}{u} - 1 + \ln(1+u)\right)$ lies between $-\frac{u}{2}\log e$ and $u \log e$. Hence

$$\frac{E_x}{\mathrm{Inf}_x} = \frac{1}{2}\log e + \sum_b w_b \log r^2 + O(m^3 4^{-m}). \tag{3}$$

Next let $B = \{b : |z_b| \leq m/4\}$. For $b \in B$ we use $\ln \binom{n}{j} = n\, h_e(j/n) - \frac{1}{2}\ln\big(2\pi j(n-j)/n\big) + O(1/n)$ for $\frac{3}{8} \leq j/n \leq \frac{5}{8}$, which follows from Robbins' form of Stirling's formula [17], where $h_e(\frac{1}{2}+\zeta) = \ln 2 - 2\zeta^2 + O(\zeta^4)$ is the binary entropy in nats. With $\zeta = z_b/(2m)$ for both $\binom{m}{b}$ and $\binom{2m}{2b}$, and $\ln(1-4\zeta^2) = O(\zeta^2)$,

$$\ln \binom{m}{b} = m\ln 2 - \frac{z_b^2}{2m} - \frac{1}{2}\ln\frac{\pi m}{2} + \varepsilon_1, \qquad \ln \binom{2m}{2b} = 2m\ln 2 - \frac{z_b^2}{m} - \frac{1}{2}\ln(\pi m) + \varepsilon_2,$$

and $\ln \binom{2m}{m} = 2m\ln 2 - \frac{1}{2}\ln(\pi m) + \varepsilon_3$, with $|\varepsilon_i| \leq C(z_b^4/m^3 + z_b^2/m^2 + 1/m)$. Therefore, for $b \in B$,

$$\log r^2 = 2m - \log\frac{\pi m}{2} + \delta_b \log e + \varepsilon_b, \qquad |\varepsilon_b| \leq C\Big(\frac{z_b^4}{m^3} + \frac{z_b^2}{m^2} + \frac{1}{m}\Big).$$

For $b \notin B$ both $\log r^2$ and $2m - \log\frac{\pi m}{2} + \delta_b \log e$ lie in $[-\log m, 6m]$, because $1 \leq r \leq \binom{m}{b}\binom{2m}{m} \leq 8^m$ and $\delta_b \leq m$. If s is binomial with parameters $2m$ and $\frac{1}{2}$, then $w_b = \Pr[s = 2b+1]$ and $z_b = s - 1 - m$, and for $m \geq 4$ Hoeffding's inequality [9, Theorem 2] gives $\sum_{b\notin B} w_b \leq \Pr[|s-m| > \frac{m}{4} - 1] \leq 2e^{-(m-4)^2/(16m)}$. Moreover $\mathbb{E}[(s-m)^4] = O(m^2)$, so $\sum_b w_b|\varepsilon_b| = O(1/m)$ over $b \in B$. Finally, $\sum_s (-1)^s \binom{2m}{s} p(s) = 0$ for every polynomial p of degree less than $2m$, so the odd and the even values of s carry equal parts of $\mathbb{E}[(s-1-m)^2] = \frac{m}{2} + 1$, and

$$\sum_b w_b \delta_b = \frac{1}{m}\cdot\frac{1}{2}\Big(\frac{m}{2} + 1\Big) = \frac{1}{4} + \frac{1}{2m}.$$

Putting these together with $\sum_b w_b = \frac{1}{2}$,

$$\sum_b w_b \log r^2 = \frac{1}{2}\Big(2m - \log\frac{\pi m}{2}\Big) + \frac{1}{4}\log e + O\Big(\frac{1}{m}\Big),$$

and by (3) $E_x/\mathrm{Inf}_x = m - \frac{1}{2}\log m + \frac{3}{4}\log e - \frac{1}{2}\log\frac{\pi}{2} + O(1/m)$. □

Remark 7. The proof shows where the weight sits: $r = 0$ exactly when $u + v$ is even, which is half of the weight, and on the other half $\log r^2$ is $2m - \log m + O(1)$ for typical b. The constant c_0 collects three contributions: the Gaussian shape of the three binomial coefficients, which gives $-\frac{1}{2}\log\frac{\pi}{2}$; their curvature, which makes $\log r^2$ grow by $\delta_b \log e$ away from the centre and contributes $\frac{1}{4}\log e$; and the gap $\Psi(r) - \log r^2 \to \log e$, which contributes $\frac{1}{2}\log e$.

Proposition 8. *In the setting of Theorem 3,*

$$\frac{E_x(\tilde{f}_k)}{\mathrm{Inf}_x(\tilde{f}_k)} = m - \frac{1}{2}\log_2 m + c_0 + \frac{5\log_2 e}{16\, m} + O\Big(\frac{1}{m^2}\Big).$$

Proof. By (3) and the tail bound in the proof of Theorem 3, it suffices to expand $\ln r$ for $|z_b| \leq m/4$ and average. Stirling's series with its first correction, $\ln n! = n\ln n - n + \frac{1}{2}\ln(2\pi n) + \frac{1}{12n} + O(n^{-3})$, gives for $\frac{3}{8} \leq j/n \leq \frac{5}{8}$

$$\ln \binom{n}{j} = n\, h_e(j/n) - \frac{1}{2}\ln\frac{2\pi j(n-j)}{n} + \frac{1}{12}\Big(\frac{1}{n} - \frac{1}{j} - \frac{1}{n-j}\Big) + O(n^{-3}).$$

Apply this to $\binom{m}{b}$, $\binom{2m}{m}$ and $\binom{2m}{2b}$ with $\zeta = z_b/(2m)$ in the first and third. The entropy terms give $m\ln 2 - m\big(h_e(\frac12+\zeta) - \ln 2\big) = m\ln 2 + 2m\zeta^2 + \frac43 m\zeta^4 + O(m\zeta^6)$, because $h_e(\frac12+\zeta) = \ln 2 - 2\zeta^2 - \frac43\zeta^4 + O(\zeta^6)$. In the logarithmic terms the factors $1-4\zeta^2$ of $\binom{m}{b}$ and $\binom{2m}{2b}$ cancel, leaving $-\frac12\ln\frac{\pi m}{2}$. The brackets of the correction terms are $-\frac{3}{m}$, $-\frac{3}{2m}$ and $-\frac{3}{2m}$ up to $O(z_b^2/m^3)$; they enter with signs $+,+,-$, so together the corrections give $-\frac{1}{4m}$. Hence

$$\ln r = m\ln 2 + \frac{z_b^2}{2m} + \frac{z_b^4}{12m^3} - \frac12\ln\frac{\pi m}{2} - \frac{1}{4m} + O\Big(\frac{z_b^2}{m^3} + \frac{z_b^6}{m^5} + \frac{1}{m^3}\Big).$$

With s binomial with parameters $2m$ and $\frac12$ as before, the alternating-sum identity gives $\sum_b w_b z_b^2 = \frac12\mathbb{E}[(s-1-m)^2] = \frac{m}{4} + \frac12$ and $\sum_b w_b z_b^4 = \frac12\mathbb{E}[(s-1-m)^4] = \frac{3m^2}{8} + O(m)$, the latter because the central fourth moment of s is $\frac{m(3m-1)}{4}$. The averages of the error terms are $O(1/m^2)$. Therefore

$$\sum_b w_b \ln r = \frac{m}{2}\ln 2 - \frac14\ln\frac{\pi m}{2} + \frac18 + \Big(\frac14 + \frac{1}{32} - \frac18\Big)\frac1m + O\Big(\frac{1}{m^2}\Big),$$

and $E_x/\mathrm{Inf}_x = \frac12\log e + \frac{2}{\ln 2}\sum_b w_b \ln r + O(m^3 4^{-m})$ gives the claim, the coefficient of $1/m$ being $\frac{2}{\ln 2}\cdot\frac{5}{32} = \frac{5}{16}\log e$. □

Remark 9. Near $P = \frac12$ the two regimes meet. Write $P = \frac12 + \varepsilon$. At $\varepsilon = 0$, $\partial_\varepsilon H_{uv} = (v-u-1)(z-1)^{u+v+1}\big(\frac{1+z}{2}\big)^{2m-u-v}$, so the terms with $u+v = 2b+1$ keep $r_{uv} = |\kappa_{2b}| + O(\varepsilon)$, while those with $u+v = 2b \ge 2$, which vanish at $\varepsilon = 0$, have

$$r_{uv} = 2\varepsilon\big|(v-u-1)\kappa_{2b} + (-1)^v\big| + O(\varepsilon^2),$$

where $v-u-1$ is odd, hence never 0. When ε is small and $\varepsilon 2^m$ is large, all these r_{uv} are large, $\Psi(r_{uv}) = \log r_{uv}^2 + \log e + o(1)$, and to first order the sum of Lemma 5 is $(k-1)\log\frac{1}{P} - \log\frac{1}{\varepsilon} - \frac12\log m + c(m)$ with $c(m)$ independent of ε; the leading term $(k-1)\log\frac{1}{P}$ differs from $2m$ by $O(m\varepsilon)$. The weight of the pairs with $u+v$ even is $\frac12 - 4^{-m}$, $v-u$ is close to normal with variance $\frac{m}{2}$ and nearly independent of $u+v$, and $\mathbb{E}\ln G^2 = -\gamma - \ln 2$ for a standard normal G, where γ is Euler's constant. With the computation in the proof of Theorem 3 for the factors κ_{2b}, this gives

$$c(m) = \log\frac{2}{\pi} + \frac{3-\gamma}{2}\log e + \frac{2}{\sqrt{\pi m}} + O\Big(\frac1m\Big), \qquad \log\frac{2}{\pi} + \frac{3-\gamma}{2}\log e = 1.096173\ldots,$$

the term $2/\sqrt{\pi m}$ coming from the lattice of odd values of $v-u-1$: for even N the sum of $\ln n^2$ over odd n with $|n| < N$ exceeds $\frac12\int_{-N}^{N}\ln x^2\,dx$ by $2\ln 2 + o(1)$, and the density of $v-u$ at 0 is about $1/\sqrt{\pi m}$. We have not made the error terms rigorous. Evaluated exactly in the first-order form, $c(m)$ is 1.423333, 1.353700, 1.313931, 1.225405 and 1.156825 at $m = 20$, 30, 40, 100 and 400, and $(c(m) - 1.096173)\sqrt{m}$ is 1.463, 1.411, 1.377, 1.292 and 1.213, approaching $2/\sqrt{\pi} = 1.128$ at the rate $1/\sqrt{m}$. For the exact sum of Lemma 5 at $k = 61$ and $\varepsilon = 10^{-4}$, 2^{-15} and 10^{-6}, the ratio minus $(k-1)\log\frac{1}{P} - \log\frac{1}{\varepsilon} - \frac12\log m$ is 1.3540, 1.3537 and 1.3537, against $c(30) = 1.353700$, and at $k = 81$ and $\varepsilon = 10^{-4}$, 10^{-6} and 2^{-20} it is 1.3142, 1.3139 and 1.3139, against $c(40) = 1.313931$. Once $\varepsilon 2^m$ is of order 1 the first-order form fails, and the ratio moves to the value of Theorem 3.

4. The tail on the unit circle

From now on $P \in (\frac12, 1)$. For $z = e^{\mathrm{i}\theta}$ and $w = \cos\theta$,

$$|z-1|^2 = 2 - 2w, \qquad |X(z)|^2 = |Y(z)|^2 = R(w) := P^2 + q^2 + 2Pqw,$$

and $X(z) = z\,\overline{Y(z)}$. Put $n_1 = u + v - 1$ and $n_2 = 2m + 1 - u - v$, so $n_1 + n_2 = 2m$. Then $|Q_{uv}(e^{\mathrm{i}\theta})| = e^{\rho(\cos\theta)}$ with

$$\rho(w) = \frac{n_1}{2}\ln(2-2w) + \frac{n_2}{2}\ln R(w), \qquad \rho''(w) = -\frac{n_1}{2(1-w)^2} - \frac{2P^2q^2n_2}{R(w)^2} \le -2P^2q^2n_2,$$

using $R \le 1$. So ρ is strictly concave on $[-1,1)$, with maximum at

$$w_0 = \frac{2Pq\,n_2 - (P^2+q^2)\,n_1}{4Pq\,m}$$

when this lies in $(-1,1)$. A short computation gives $2-2w_0 = n_1/(2Pqm)$ and $R(w_0) = n_2/(2m)$, hence

$$\rho(w_0) = \frac{n_1}{2}\ln\frac{n_1}{2Pqm} + \frac{n_2}{2}\ln\frac{n_2}{2m}. \tag{4}$$

The law β_u has mean mq and variance mPq. For $\lambda \ge 1$ let W_λ be the set of (u,v) with $|u - mq| \le \lambda\sqrt{m}$ and $|v - mq| \le \lambda\sqrt{m}$. Below, c, C are positive constants that may depend on P and λ but not on m or on $(u,v) \in W_\lambda$, and m is large enough for each statement. On W_λ, $n_1 = 2mq + O(\sqrt{m})$ and $w_0 = w^* + O(m^{-1/2})$, where

$$w^* = \frac{2P-1}{2P} = \cos\alpha_P \in (0, \tfrac{1}{2}).$$

Put $\theta_0 = \arccos w_0$, $z_0 = e^{\mathrm{i}\theta_0}$, $\theta^* = \alpha_P$ and $z^* = e^{\mathrm{i}\theta^*}$. Since $1 - w^* = 1/(2P)$, the real part of $(z^*-1)(q + P\bar{z}^*)$ is $(P-q)(1-w^*) = w^*$ and its imaginary part is $\sin\theta^*$, and $R(w^*) = P^2 + q^2 + q(P-q) = P$; so

$$(z^*-1)\,\overline{X(z^*)} = z^*, \qquad |X(z^*)|^2 = P, \qquad Y(z^*) = z^*\,\overline{X(z^*)}. \tag{5}$$

Lemma 10. *On W_λ, $\rho(w_0) + (n_2-1)\ln\frac{1}{P} \ge (m-1)\ln\frac{1}{P}$.*

Proof. Regard the left side, by (4), as a function $G(n_1)$ of a real $n_1 \in (0, 2m)$ with $n_2 = 2m - n_1$. Then $G'(n_1) = \frac{1}{2}\ln\frac{n_1}{Pq\,n_2} + \ln P$, which vanishes at $n_1 = 2mq$, and $G'' = \frac{1}{2n_1} + \frac{1}{2n_2} > 0$. So $G \ge G(2mq) = mq\ln\frac{1}{P} + mP\ln P + (2mP-1)\ln\frac{1}{P} = (m-1)\ln\frac{1}{P}$. □

Fix $(u,v) \in W_\lambda$ and let $\Lambda(\theta) = \ln Q_{uv}(e^{\mathrm{i}\theta}) - \mathrm{i}m\theta$ for complex θ with $|\theta - \theta^*| \le \delta$, where $\delta > 0$ is so small that $e^{\mathrm{i}\theta}$ stays away from 1, $-q/P$ and $-P/q$, the zeros of $z-1$, X and Y; the logarithm is the branch continuous in this disc. Then $|\Lambda''|, |\Lambda'''| \le Cm$ on the disc. For real θ, $\operatorname{Re}\Lambda(\theta) = \rho(\cos\theta)$, so $\operatorname{Re}\Lambda'(\theta_0) = 0$ and $\operatorname{Re}\Lambda''(\theta_0) = \rho''(w_0)\sin^2\theta_0 \le -cm$. Put

$$b = \operatorname{Im}\Lambda'(\theta_0), \qquad A = -\Lambda''(\theta_0), \qquad T_{uv} = \frac{1}{\sqrt{2\pi A}}\exp\Big(\Lambda(\theta_0) - \frac{b^2}{2A}\Big).$$

Lemma 11. *On W_λ: $|b| \le C\sqrt{m}$, $|T_{uv}| \ge c\,m^{-1/2}e^{\rho(w_0)}$, and $t_{uv} = 2\operatorname{Re}T_{uv} + O(|T_{uv}|\,m^{-1/5})$.*

Proof. With $a = u/m$ and $a' = v/m$, $\Lambda'(\theta)/m$ differs by $O(1/m)$ from

$$\mathrm{i}z\Big(\frac{a+a'}{z-1} + \frac{(1-a)P}{X(z)} + \frac{(1-a')q}{Y(z)}\Big) - \mathrm{i}, \qquad z = e^{\mathrm{i}\theta},$$

a smooth function of (a, a', θ). It vanishes at $a = a' = q$, $\theta = \theta^*$: by (5), $1/(z^*-1) = \overline{X^*}/z^*$, $1/X^* = \overline{X^*}/P$ and $1/Y^* = X^*/(Pz^*)$, where $X^* = X(z^*)$, so

$$z^*\Big(\frac{2q}{z^*-1} + \frac{P^2}{X^*} + \frac{Pq}{Y^*}\Big) = 2q\overline{X^*} + Pz^*\overline{X^*} + qX^* = 3q^2 + P^2 + 4Pq\,w^* = 1.$$

The rest is the proof of [24, Lemma 5]. Since $a-q$, $a'-q$ and $\theta_0 - \theta^*$ are $O(m^{-1/2})$, $|\Lambda'(\theta_0)| \le C\sqrt{m}$, and $\operatorname{Re}(b^2/A) = b^2\operatorname{Re}A/|A|^2 \le C$, which gives the lower bound on $|T_{uv}|$. By Cauchy's

formula [5, Chapter VIII], $t_{uv} = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{\Lambda(\theta)}d\theta$. On $J_+ = \{|\theta-\theta_0| \le m^{-2/5}\}$, Taylor's formula $\Lambda(\theta_0+h) = \Lambda(\theta_0) + \mathrm{i}bh - \frac{1}{2}Ah^2 + O(m|h|^3)$ and a Gaussian integral give $T_{uv} + O(|T_{uv}|m^{-1/5})$; the integral over $J_- = -J_+$ is its complex conjugate, as Q_{uv} has real coefficients. Off $J_+ \cup J_-$, $|\cos\theta - w_0| \ge c|\theta \mp \theta_0|$ because θ_0 stays away from 0 and π, and $\rho'' \le -cm$ gives a modulus at most $e^{\rho(w_0)-cm^{1/5}}$. □

Lemma 12. *If (u,v) and $(u+1,v)$ lie in W_λ, then $\arg T_{u+1,v} - \arg T_{uv} = \alpha_P + O(m^{-1/2})$ modulo 2π.*

Proof. Let $g(z) = (z-1)/X(z)$, so $Q_{u+1,v} = gQ_{uv}$. Passing from (u,v) to $(u+1,v)$ changes w_0 by $-1/(4Pqm)$, so θ_0 moves by $O(1/m)$, and as in [24, Lemma 6] $\arg T$ moves by $\operatorname{Im}\ln g(z_0) + O(m^{-1/2})$. By (5), $g(z^*) = (z^*-1)\overline{X^*}/|X^*|^2 = z^*/P$, so $\arg g(z^*) = \theta^* = \alpha_P$, and $z_0 - z^* = O(m^{-1/2})$. □

If P is rational then so is $\cos\alpha_P = (2P-1)/(2P)$, and it lies in $(0,\frac{1}{2})$; by Niven's theorem [21] a rational multiple of π with rational cosine has cosine in $\{0, \pm\frac{1}{2}, \pm 1\}$, so α_P/π is irrational. This is the last claim of Theorem 1.

5. Proof of Theorem 1

The influence is Lemma 5 and the upper bound is (1). The lower bound is the argument of [24, Section 4], which we repeat with the constants of a general P. Fix $\varepsilon \in (0,\frac{1}{4})$, $\lambda \ge 1$ and $L \ge 1$ so large that for every real ψ_0 at most $3\varepsilon L$ of the numbers $\ell = 0,\dots,L-1$ satisfy $|\cos(\psi_0 + \ell\alpha_P)| < 2\varepsilon$; this is possible because $\ell\alpha_P/\pi$ is equidistributed modulo 1 [20]. Fix v and cut the u with $(u,v) \in W_\lambda$ into blocks of L consecutive values. By Lemma 12, in each block at most $3\varepsilon L$ values have $|\cos\arg T_{uv}| < \varepsilon$ once m is large. On W_λ, $\beta_{u+1}/\beta_u = (m-u)q/((u+1)P) = 1 + O(m^{-1/2})$, so the weights in a block differ by a factor at most 2, and the bad values and the left-over values carry $\beta_u\beta_v$-weight at most $6\varepsilon + o(1)$ in all.

Let $(u,v) \in W_\lambda$ have $|\cos\arg T_{uv}| \ge \varepsilon$. By Lemma 11, $|t_{uv}| \ge \varepsilon|T_{uv}| \ge c\varepsilon m^{-1/2}e^{\rho(w_0)}$, and with $2m-u-v = n_2 - 1$ and Lemma 10,

$$r_{uv} \ge \frac{2|t_{uv}|}{P^{n_2-1}} - (P-q) \ge c\varepsilon m^{-1/2}P^{-(m-1)} - 1,$$

so $\Psi(r_{uv}) \ge \log r_{uv}^2 \ge 2m\log\frac{1}{P} - \log m - C_\varepsilon$. By Chebyshev's inequality W_λ has weight at least $(1 - Pq/\lambda^2)^2$, and since $\Psi \ge 0$, Lemma 5 gives

$$\frac{E_x^P}{\mathrm{Inf}_x^P} \ge \Big(\big(1-\tfrac{Pq}{\lambda^2}\big)^2 - 6\varepsilon - o(1)\Big)\Big(2m\log\frac{1}{P} - \log m - C_\varepsilon\Big).$$

Dividing by $\log(1/\mathrm{Inf}_x^P) = 2m\log\frac{1}{P}$ and letting $m \to \infty$, then $\lambda \to \infty$ and $\varepsilon \to 0$, gives $\liminf \ge 1$. □

6. Numerical checks

Lemma 5 reduces the ratio to $(m+1)^2$ terms whose ingredients are integers over powers of the denominator of P; the code computes them exactly in integer arithmetic and evaluates each term of Φ as $p\log(1+q/p)$ with a $\log(1+\cdot)$ routine. For $t=1$ the values agree with those of [24], and for $t = 0,1,2,3$ at $k \le 27$ with a rational-arithmetic implementation. The quotient of $E_x(F_{k,t})/\mathrm{Inf}_x(F_{k,t})$ by $(k-1)\log_2(1/P_t)$, with $P_2 = \frac{11}{16}$, $P_3 = \frac{21}{32}$ and $P_4 = \frac{163}{256}$, is:

k	41	81	121	161	201	251	301
$t=2$	0.849	0.899	0.922	0.936	0.946	0.954	0.960
$t=3$	0.854	0.902	0.928	0.941	0.950	0.958	0.964
$t=4$	0.858	0.909	0.931	0.944	0.953	0.960	0.966

For $\tilde{f}_k$ the formula of Theorem 3 agrees with the double sum at $k = 13$, 41, 81 and 121 to six decimals, and the identity of Lemma 6 was checked against direct expansion for $m \le 11$. The quotient by $k-1$ is 0.466, 0.476, 0.482, 0.490, 0.496 and 0.499 at $k = 41$, 81, 121, 301, 1001 and 4001, and $E_x/\mathrm{Inf}_x - m + \frac{1}{2}\log_2 m$ is 0.7796, 0.7677, 0.7639, 0.7593, 0.7572 and 0.7565 at the same values of k. The difference from $c_0 = 0.75627$, multiplied by m, is 0.467, 0.459, 0.456, 0.453, 0.451 and 0.4510 at these k, and 0.4509 at $k = 8001$, in line with the error term $O(1/m)$ of Theorem 3. Extrapolating linearly in $1/m$ between consecutive values of k from 1001 to 8001 gives 0.450842 each time, which is $\frac{5}{16}\log_2 e$ to six places, as in Proposition 8. After that term is removed, the remainder multiplied by m^2 is 0.322, 0.311, 0.307, 0.303, 0.301, 0.301 and 0.301 at $k = 41, 81, 121, 301, 1001, 2001, 4001$.

Acknowledgements and funding

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [25] and working together inside the Brain framework [27], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [26], the Math Brain repository [25] and the Brain framework repository [27].

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [27] and Math Brain [25]. The family $F_{k,t}$ is from [23] and the method for $P = \frac{3}{4}$ from [24]. Claude Opus 5.5 (Anthropic) found Theorems 1 and 3 and their proofs, wrote the numerical checks and prepared the manuscript. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-forced-coordinate-bias`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [25].

References

[1] S. Arunachalam, S. Chakraborty, M. Koucký, N. Saurabh, R. de Wolf, Improved bounds on Fourier entropy and min-entropy, ACM Trans. Comput. Theory 13 (2021), no. 4, 1–40, doi:10.1145/3470860; arXiv:1809.09819.

[2] G. Shalev, On the Fourier entropy influence conjecture for extremal classes, arXiv:1806.03646, 2018.

[3] F. Chang, A lower bound for the Fourier entropy of Boolean functions on the biased hypercube, arXiv:2511.07739, 2025.
[4] S. Chakraborty, R. Kulkarni, S. V. Lokam, N. Saurabh, Upper bounds on Fourier entropy, Theoret. Comput. Sci. 654 (2016), 92–112.
[5] P. Flajolet, R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
[6] M. J. González, P. MacManus, M. C. Pereyra, Further evidence towards the Fourier entropy-influence conjecture, arXiv:2606.00246, 2026.
[7] E. Friedgut, G. Kalai, Every monotone graph property has a sharp threshold, Proc. Amer. Math. Soc. 124 (1996), 2993–3002.
[8] X. Han, A new bound for the Fourier-entropy-influence conjecture, Combinatorica 45 (2025), no. 1, Art. 4, doi:10.1007/s00493-024-00133-z; arXiv:2312.08271.
[9] W. Hoeffding, Probability inequalities for sums of bounded random variables, J. Amer. Statist. Assoc. 58 (1963), 13–30.
[10] R. Hod, Improved lower bounds for the Fourier entropy/influence conjecture via lexicographic functions, arXiv:1711.00762, 2017.
[11] E. Kelman, G. Kindler, N. Lifshitz, D. Minzer, M. Safra, Towards a proof of the Fourier-entropy conjecture?, Geom. Funct. Anal. 30 (2020), 1097–1138.
[12] N. Keller, E. Mossel, T. Schlank, A note on the entropy/influence conjecture, Discrete Math. 312 (2012), 3364–3372, arXiv:1105.2651.
[13] F. J. MacWilliams, N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland, 1977.
[14] R. O'Donnell, *Analysis of Boolean Functions*, Cambridge University Press, 2014.
[15] R. O'Donnell, L.-Y. Tan, A composition theorem for the Fourier entropy-influence conjecture, ICALP 2013, Lecture Notes in Comput. Sci. 7965, 780–791, doi:10.1007/978-3-642-39206-1_66.
[16] R. O'Donnell, J. Wright, Y. Zhou, The Fourier entropy-influence conjecture for certain classes of Boolean functions, ICALP 2011, Lecture Notes in Comput. Sci. 6755, 330–341.
[17] H. Robbins, A remark on Stirling's formula, Amer. Math. Monthly 62 (1955), 26–29.
[18] S. Verdú, T. Weissman, The information lost in erasures, IEEE Trans. Inform. Theory 54 (2008), 5030–5058.
[19] A. Wan, J. Wright, C. Wu, Decision trees, protocols, and the Fourier entropy-influence conjecture, ITCS 2014, arXiv:1312.3003.
[20] H. Weyl, Über die Gleichverteilung von Zahlen mod. Eins, Math. Ann. 77 (1916), 313–352.
[21] I. Niven, *Irrational Numbers*, Carus Mathematical Monographs 11, Mathematical Association of America, 1956, Corollary 3.12.
[22] Y. Shmalo, The one-coordinate step in the Fourier entropy-influence conjecture, `https://github.com/yspennstate/fourier-entropy-influence-one-coordinate`, 2026.
[23] Y. Shmalo, Hidden coordinates and selection rules in the Fourier entropy-influence conjecture, `https://github.com/yspennstate/fei-hidden-coordinates`, 2026.
[24] Y. Shmalo, A forced coordinate with nearly maximal erasure entropy, `https://github.com/yspennstate/fei-forced-coordinate`, 2026.
[25] Y. Shmalo, *Math Brain: public snapshot, code and tests*, 2026, `https://github.com/yspennstate/math-brain-public`.
[26] Y. Shmalo, *Math Brain: shared, auditable memory for multi-agent mathematical research*, October 2026, `https://github.com/yspennstate/math-brain-public/tree/main/papers/math-brain-2026-10-24`.
[27] Y. Shmalo, *Brain Framework*, public software repository, 2026, `https://github.com/yspennstate/brain-framework-public`.